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ME 3902 Test 3 Task E: Reading Summary

I2C

The Inter-Integrated Circuit (I2C) protocol is a synchronous serial communication protocol designed to let multiple “peripheral” chips communicate with one or more “controller” chips over short distances within a single device [1]. Unlike SPI, which needs four or more wires, I2C only requires two signal lines: SDA, which carries the data, and SCL, which carries a clock signal shared by every device on the bus [1]. Because the clock is physically shared, devices do not need to agree on a data rate ahead of time the way asynchronous serial devices do. Each peripheral is assigned a 7-bit address, and the controller begins every message with the address of the device it wants to talk to, which allows many devices (like the MPU6050 IMU at its default address of 0x68) to share the same two wires [1]. The bus lines are open-drain and require pull-up resistors, meaning devices can only pull the lines low, and the receiver acknowledges each byte of a message with an ACK/NACK bit [1]. Standard I2C runs at 100 kHz with a “fast mode” at 400 kHz – slower than SPI, but far more flexible when connecting several sensors to one microcontroller [1].

Baud Rate

Baud rate applies to asynchronous serial communication (UART), where there is no shared clock line, so both devices must agree ahead of time on how fast data will be sent [2]. The baud rate specifies how fast data is sent over a serial line, usually expressed in bits per second, and inverting it gives the time required to transmit a single bit – this is what tells the transmitter how long to hold the line high or low and tells the receiver when to sample it [2]. Common standard rates include 1200, 2400, 4800, 9600, 19200, 38400, 57600, and 115200 bps, with 9600 being typical when speed isn’t critical [2]. Since each frame adds a start and stop bit, 10 bits of transmission are needed for every 8 bits of actual data, cutting into real throughput [2]. The critical requirement is that both devices operate at the same rate: if data is transmitted at 9600 bps but received at 19200 bps, the receiver samples the line at the wrong times and outputs garbage characters [2].

LiDAR Using the Arduino Nano, Uno, and Mega

These two concepts come together when selecting a microcontroller for a LiDAR sensor like the TFMini, a time-of-flight sensor that measures 30 cm to 12 m and streams data over UART at 115200 baud by default [3]. The Arduino Uno and Nano each have only one hardware UART, and it is already wired to the USB-to-serial converter used for uploading code and the Serial Monitor, so the LiDAR must instead use the SoftwareSerial library, which “bit-bangs” serial in software – an approach that is processor-intensive and less precise than a true UART [2]. SparkFun notes that an 8 MHz board can only handle about 57600 baud reliably in software serial, and even a 16 MHz Uno or Nano running 115200 through SoftwareSerial can produce unreliable readings [3]. The Arduino Mega solves this with four hardware UARTs (Serial through Serial3), so a LiDAR – or several – can each run at full speed on dedicated hardware serial ports while USB debugging stays free, and its 8 KB of SRAM (versus 2 KB on the Uno/Nano) leaves room for the data buffers. My selection would therefore depend on the system: a single LiDAR at a reduced baud rate suits the Nano where space is tight, the Uno works for simple bench prototyping, and multiple LiDARs streaming at full rate require the Mega.

Works Cited

[1] SparkFun Electronics, "I2C," SparkFun Learn. [Online]. Available: https://learn.sparkfun.com/tutorials/i2c/all. [Accessed 4 July 2026].

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[2] SparkFun Electronics, "Serial Communication," SparkFun Learn. [Online]. Available: https://learn.sparkfun.com/tutorials/serial-communication/all. [Accessed 4 July 2026].

[3] SparkFun Electronics, "TFMini - Micro LiDAR Module Hookup Guide," SparkFun Learn. [Online]. Available: https://learn.sparkfun.com/tutorials/tfmini---micro-lidar-module-hookup-guide/all. [Accessed 4 July 2026].